Rescoring-LLM Comparison for Intracortical Brain-to-Text Decoding

Analysis of validation WER/CER across five candidate LLMs used for zero-shot rescoring on top of CTC beam search + n-gram (KenLM) fusion

Executive Summary

Five LLMs were evaluated as zero-shot rescorers on top of a CTC beam-search decoder fused with a KenLM n-gram language model. For each model, the n-gram weight (lm_weight) was swept from 0.5 to 6.0, with the LLM-fusion weight (λ, lambda) re-tuned at every point. **Qwen2.5-7B produced the best overall result (WER 7.26%, CER 5.18%)**, narrowly ahead of the much smaller Llama3.2-3B-Instruct (WER 7.33%, CER 5.16% — the lowest CER of any model). Qwen3-4B and Gemma3-4B behave differently from the other three: across the sweep both do show a nonzero, benefit-giving λ while the n-gram weight is still low, but that contribution fades to λ = 0 once lm_weight is pushed up into the range that gives each model its lowest overall WER. For Qwen2.5-7B, Llama3.2-3B-Instruct, and Mistral-7B, by contrast, a nonzero λ persists even at the best configuration. Mistral-7B landed in the middle of the pack on absolute error.

Rescoring LLM	Best lm_weight	Best λ (lambda)	WER (%)	CER (%)	LLM contributes?
Qwen2.5-7B	5.5	1.25	7.26	5.18	Yes
Llama3.2-3B-Instruct	5.5	0.5	7.33	5.16	Yes
Qwen3-4B	5.5	0.0	7.37	5.33	Yes*
Mistral-7B	4.0	0.5	8.27	5.75	Yes
Gemma3-4B	4.0	0.0	8.33	5.87	Yes*

Table 1. Best validation operating point per rescoring LLM (green row = overall best WER/CER). * Qwen3-4B and Gemma3-4B do show λ > 0 (a real benefit) at lower lm_weight values earlier in the sweep, but λ settles to 0 at the specific best-WER row shown here.

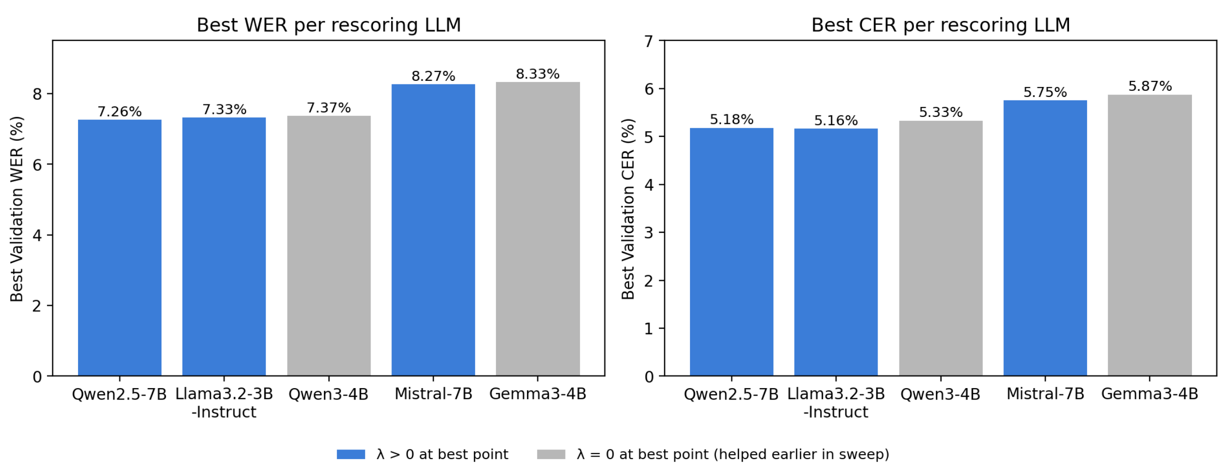

Figure 1. Best WER and CER achieved per rescoring LLM. Grey bars mark models where the jointly optimal λ at the best operating point is 0 — both still show λ > 0 (and a real benefit) at lower n-gram weights earlier in the sweep.

How Error Rate Responds to n-gram Weight

Across all five models the shape of the curve is the same: WER falls steeply as lm_weight increases from 0.5 to about 2.5–3.0, then flattens into a shallow plateau from lm_weight ≈ 4 onward, with the optimum sitting at lm_weight = 4.0–5.5 for every model. This indicates the n-gram LM is doing the bulk of the error-reduction work, and the marginal value of pushing lm_weight higher still (beyond ~5.5–6.0) is essentially zero or slightly negative — all five curves tick back up at lm_weight = 6.0.

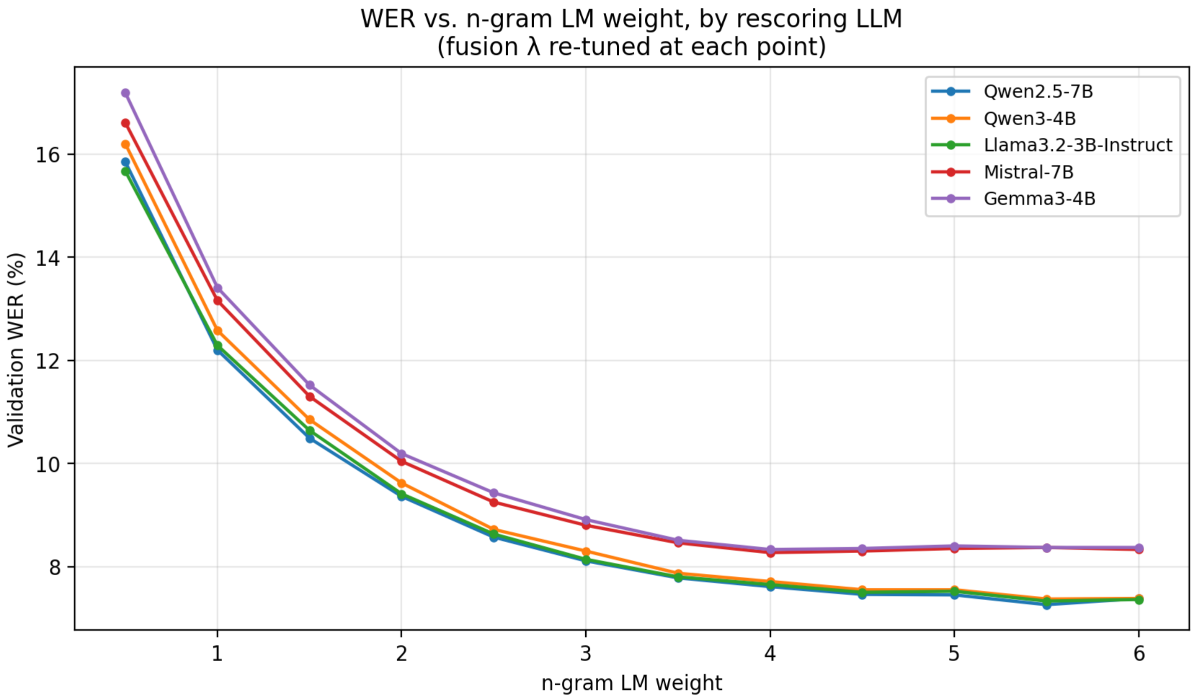

Figure 2. Validation WER vs. n-gram LM weight for each rescoring LLM (λ re-tuned at every point).

The two strongest curves (Qwen2.5-7B and Llama3.2-3B-Instruct) sit essentially on top of each other for lm_weight ≥ 3.5, while Mistral-7B and Gemma3-4B plateau roughly one point higher in WER — suggesting the gap between the two model clusters is set early (by lm_weight ≈ 2.0) and persists rather than closing as more n-gram weight is added.

Per-Model Observations

Qwen2.5-7B — best overall (WER 7.26%, CER 5.18%)

The only model whose best λ is a clear, substantial 1.25, meaning it draws the most on LLM rescoring of the five candidates, and that contribution translates into the lowest WER and CER in the comparison. The before/after fusion chart for this model shows a modest but real gain from n-gram-only (WER 7.37% → 7.26%; CER 5.33% → 5.18%), a roughly 1.5% relative WER reduction and 2.8% relative CER reduction from adding the LLM.

Llama3.2-3B-Instruct — best CER (5.16%), near-best WER (7.33%)

Despite being roughly 2–3x smaller than Qwen2.5-7B and Mistral-7B, this model is statistically competitive with Qwen2.5-7B on WER and edges it out on CER. Its λ of 0.5 shows a real but more modest LLM contribution. Given the much lower inference cost of a 3B model, this is arguably the most attractive option if compute/latency matters for the final pipeline, at a cost of ~0.07 WER points relative to Qwen2.5-7B.

Qwen3-4B — helps early, drops out at the optimum (WER 7.37%)

λ is nonzero (0.75–2.0) for lm_weight 0.5 through 3.0, meaning LLM rescoring genuinely reduces error while the n-gram model is still under-weighted. From lm_weight 3.5 onward — including the best point, 5.5 — the re-tuned λ settles at 0.0, so at the operating point that gives the lowest overall WER the n-gram-only decode already matches the best LLM-assisted result. The reported final validation numbers (WER and CER identical before/after LLM fusion, both 7.37%/5.33%) reflect that specific comparison at the best point — they don't mean the LLM never helped anywhere in the sweep.

Gemma3-4B — intermittent LLM benefit, weakest overall result (WER 8.33%)

λ is nonzero at lm_weight 0.5, 1.0, 2.0, and 2.5 (up to 1.25), so LLM fusion does contribute at those points — it's just less consistent than for Qwen3-4B (zero already appears at lm_weight 1.5, ahead of the final drop-out at 3.0+). As with Qwen3-4B, the best operating point (lm_weight 4.0) lands on λ = 0, and Gemma3-4B also has the worst n-gram-only floor of the five candidates, making it the weakest overall regardless of the LLM-fusion question.

Mistral-7B — mid-pack, but the most fusion-hungry model that still helps (WER 8.27%)

λ = 0.5 shows Mistral-7B does draw on LLM rescoring, but its n-gram-only floor is already higher than the top two models, so it can't close the gap. Its optimum also sits at a lower lm_weight (4.0) than the 5.5 favored by the top three models.

Data-Quality Note Worth Checking Before Using These Numbers

One “Final validation numbers / Overall metrics” block in the source file is internally inconsistent with its own surrounding section and is worth re-checking against your raw logs before it goes into the paper:

- It reports **Beam+NGram = 11.52% WER / 7.65% CER**, which matches Gemma3-4B's row at lm_weight = 1.5, $\lambda = 0.0$ exactly — not a number that appears anywhere in the Mistral-7B sweep it is printed next to.
- It reports **Beam+NGram+LLM = 8.27% WER / 5.75% CER**, which matches Mistral-7B's own best row (lm_weight = 4.0, $\lambda = 0.5$) exactly.
- In other words, this summary block appears to pair a Gemma3-4B n-gram-only baseline with a Mistral-7B LLM-fused result, most likely a page/figure re-ordering artifact from PDF export rather than a real comparison. The PER of 12.12% is consistent across every block and is not affected by this, since PER is computed before any LM is applied.

Recommendation: regenerate this specific panel directly from the run logs for whichever model it is meant to represent before citing the 11.52% → 8.27% improvement figure in the paper, since as it stands it compares two different models' configurations rather than one model's before/after.

Recommendation for the Pipeline / Paper

- **Primary result:** report Qwen2.5-7B as the main rescoring LLM (lm_weight = 5.5, $\lambda = 1.25$) — best WER (7.26%) and CER (5.18%) of all candidates tested.
- **Efficiency ablation:** Llama3.2-3B-Instruct is worth featuring as a lightweight alternative — it is within 0.1 WER point of Qwen2.5-7B and actually has the best CER, at roughly half the parameter count.
- **Worth a sentence in Discussion:** Qwen3-4B and Gemma3-4B do contribute at lower n-gram weights, but that contribution vanishes at the n-gram weight that minimizes overall WER for each — unlike the other three models, where fusion still helps even at the best configuration. That's a genuine, citable pattern: LLM rescoring's value here seems to depend on how much headroom the n-gram model leaves, not just on which LLM is used.
- Fix the mismatched Beam+NGram/+LLM summary panel noted above before it is used as a headline pipeline number.

Appendix: Full Sweep Results, All Models

Complete lm_weight × λ (lambda) sweep for every rescoring LLM, as reported in the source file. λ was re-tuned at each lm_weight point; the row achieving the lowest WER for each model is highlighted and marked ’best’.

Qwen2.5-7B

lm_weight	λ (lambda)	WER (%)	CER (%)	
0.5	2.50	15.85	10.13	
1.0	2.50	12.20	8.00	
1.5	2.50	10.49	7.04	
2.0	1.75	9.36	6.30	
2.5	1.75	8.57	5.86	
3.0	1.50	8.11	5.58	
3.5	1.25	7.78	5.42	
4.0	0.50	7.61	5.31	
4.5	0.50	7.46	5.23	
5.0	1.25	7.45	5.24	
5.5	**1.25**	**7.26**	**5.18**	**← best**
6.0	0.00	7.38	5.40	

Qwen3-4B

lm_weight	λ (lambda)	WER (%)	CER (%)	
0.5	2.00	16.20	10.22	
1.0	2.00	12.58	8.23	
1.5	1.25	10.85	7.18	
2.0	1.50	9.62	6.48	
2.5	0.75	8.72	5.94	
3.0	1.25	8.30	5.71	
3.5	0.00	7.87	5.46	
4.0	0.00	7.71	5.43	
4.5	0.00	7.55	5.38	
5.0	0.00	7.55	5.39	
5.5	**0.00**	**7.37**	**5.33**	**← best**
6.0	0.00	7.38	5.40	

Llama3.2-3B-Instruct

lm_weight	λ (lambda)	WER (%)	CER (%)	
0.5	2.50	15.67	9.97	
1.0	1.50	12.29	8.02	
1.5	1.00	10.64	7.07	
2.0	1.00	9.41	6.33	
2.5	1.25	8.63	5.86	
3.0	1.00	8.14	5.60	
3.5	1.00	7.80	5.42	
4.0	0.75	7.65	5.31	
4.5	0.75	7.50	5.25	
5.0	0.75	7.52	5.22	
5.5	**0.50**	**7.33**	**5.16**	**← best**
6.0	0.50	7.36	5.25	

Mistral-7B

lm_weight	λ (lambda)	WER (%)	CER (%)	
0.5	1.25	16.61	10.47	
1.0	1.25	13.16	8.57	
1.5	1.25	11.30	7.45	
2.0	0.75	10.04	6.78	
2.5	0.75	9.25	6.28	
3.0	1.25	8.80	6.08	
3.5	0.75	8.46	5.86	
4.0	**0.50**	**8.27**	**5.75**	**← best**
4.5	0.75	8.30	5.83	
5.0	0.50	8.35	5.91	
5.5	0.00	8.37	6.06	
6.0	0.25	8.33	5.98	

Gemma3-4B

lm_weight	λ (lambda)	WER (%)	CER (%)	
0.5	1.25	17.19	10.78	
1.0	1.25	13.41	8.72	
1.5	0.00	11.52	7.65	
2.0	0.75	10.19	6.85	
2.5	0.25	9.43	6.34	
3.0	0.00	8.91	6.15	
3.5	0.00	8.51	5.92	
4.0	**0.00**	**8.33**	**5.87**	**← best**
4.5	0.00	8.35	5.90	
5.0	0.00	8.40	6.02	
5.5	0.00	8.37	6.06	
6.0	0.00	8.37	6.10	